

Explaining the MeanFlow Equations

Speaker bullets for presentation slides 30–34

The five equations on slides 30–34 build a single argument: *define* the object we want → show *why* it is useful → expose the *catch* → make it *computable* → *train* a network on it. The three-panel figure (Diffusion ≈ 35 calls / FlowCast ≈ 10 / MeanFlow 1) stays fixed while the equation underneath advances.

Slide 30 — Average velocity (definition)

$$u(z_r, r, t) \triangleq \frac{1}{t-r} \int_r^t v(z_\tau, \tau) d\tau$$

- **The picture on the right is the whole story:** diffusion curves to the forecast in ≈ 35 calls, FlowCast straightens it to ≈ 10 , MeanFlow wants to jump there in **1**.
- $v(z_\tau, \tau)$ is the **instantaneous** velocity FlowCast already learns — the local tangent at each point of the path.
- u is its **time-average over the interval** $[r, t]$: displacement divided by interval length. It is the single constant velocity that carries you from z_r to z_t .
- Crucially, u is a **property of the flow itself**, not of any network — and as $t \rightarrow r$ it collapses back to v (one evaluation = FlowCast).

Slide 31 — Single-step transport (why we want u)

$$z_t = z_r + (t-r)u(z_r, r, t)$$

- This is just the definition rearranged — but read it as a **sampler**: knowing u turns transport into **one algebraic step**, no ODE solver.
- Set $(r, t) = (0, 1)$ and a **single evaluation maps noise** → **forecast** (the green dot, one call).
- This is where MeanFlow differs from FlowCast in kind, not degree: FlowCast *integrates* many small steps; MeanFlow takes **one exact step**.
- **The catch:** u is defined by an intractable path integral (slide 30), so we cannot regress a network onto it directly. The cost has moved from inference into training.

Slide 32 — The MeanFlow identity (boxed = key result)

$$u(z_r, r, t) = v(z_r, r) + (t-r) \frac{d}{dr} u(z_r, r, t)$$

- We **kill the integral by differentiating instead of integrating**. Multiply slide 30 by $(t-r)$, differentiate in the state-time r (product rule + fundamental theorem of calculus).

- Result relates the field we **want** (u) to the field we **can sample** (v), using only quantities at a **single point of a single trajectory** — the integral is gone.
- The term $(t - r) \frac{d}{dr} u$ is the **correction** that lifts the local velocity v to the interval average u .
- At $t = r$ it reduces to $u = v$ — **plain flow matching is the boundary condition**, which is exactly why MeanFlow is a clean generalisation of FlowCast.

Slide 33 — Expanding the total derivative

$$\frac{d}{dr} u(z_r, r, t) = v(z_r, r) \partial_z u + \partial_r u$$

- The $\frac{d}{dr}$ in the identity is a **total** derivative — the state z_r itself moves along the trajectory as r changes ($dz_r/dr = v$).
- Chain rule splits it into two **partials we can compute**: motion through space ($v \partial_z u$) plus the explicit dependence on the state-time ($\partial_r u$). (t is held fixed, so no ∂_t term.)
- This is the form that maps directly onto a **Jacobian–vector product (JVP)**: one extra forward-mode pass with tangent $(v, 1, 0)$ for (z, r, t) — no higher-order gradients.
- In short: this line is what makes the identity **actually trainable** on a network.

Slide 34 — Training objective

$$\mathcal{L}_{\text{MF}} = \mathbb{E} \left[\text{sg}(w) \left\| u_\theta(z_r, r, t, c_t) - \text{sg}(u_{\text{tgt}}) \right\|_{2,\beta}^2 \right] + \lambda_{\text{spec}} \mathcal{L}_{\text{spec}},$$

$$u_{\text{tgt}} = v + (t - r)(v \partial_z u_\theta + \partial_r u_\theta)$$

- We **regress the network onto the identity’s right-hand side**: the target u_{tgt} is just slide 32 with slide 33 substituted in, and the network **bootstraps its own derivative**.
- $v = x_1 - x_0$ is the conditional I-CFM velocity (same substitution FlowCast makes, valid in expectation) — **still teacher-free, no diffusion model queried**.
- **stop-gradient (sg)** on the target and on the weight: we do not backprop through the JVP, so training stays first-order and stable.
- $w = (\|\Delta\|_{2,\beta}^2 + \varepsilon)^{-p}$ is adaptive loss weighting (down-weights hard samples); $\|\cdot\|_{2,\beta}$ carries the same per-channel weights $\beta = (1, 1, 1, 2)$ and $\mathcal{L}_{\text{spec}}$ the same **qppepre** spectral term as FlowCast.
- A network at zero loss **satisfies the identity $\rightarrow$ equals the true average velocity**. And on the 75% of the batch where $t = r$, $u_{\text{tgt}} = v$ and this is **literally the FlowCast loss** — so MeanFlow vs FlowCast is an A/B on the objective alone.

Presentation note. The same three-panel figure repeats on all five slides while the equation underneath advances. If that is intentional (equations animating in over a fixed visual), these bullets match it. If you prefer, let the coloured dots / trajectory in the figure light up in step with the equation — e.g. highlight the single green MeanFlow dot on slide 31 when you

introduce one-step transport.